BOUNDED DERIVED CATEGORY OF A COUNTABLE ABELIAN CATEGORY IS COUNTABLE

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Editorial note. Source proof ends mid-sentence (the final line reads "as a countable colimit of" and stops); the missing tail is the grammatical object (countable sets); transcribed as-is without completion.

Allowed standard prerequisites (editorial).

- Derived Hom colimit. derived-lemma-bounded-derived (cross-chapter, named). Presents Hom in D^b as a filtered colimit over quasi-isomorphisms.

Source reference. Original reference 05RW: focused-seeds/geometry/S9-eqv-countable-abelian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be a countable abelian category. Then $D^b(\mathcal{A})$ is countable.*

Proof. It suffices to prove the statement for $D(\mathcal{A})$ as the others are full subcategories of this one. Since every object in $D(\mathcal{A})$ is a complex of objects of $\mathcal{A}$ it is immediate that the set of isomorphism classes of objects of $D^b(\mathcal{A})$ is countable. Moreover, for bounded complexes $A^\bullet$ and $B^\bullet$ of $\mathcal{A}$ it is clear that $\mathrm{Hom}_{K^b(\mathcal{A})}(A^\bullet, B^\bullet)$ is countable. We have

$$\mathrm{Hom}_{D^b(\mathcal{A})}(A^\bullet, B^\bullet) = \mathrm{colim}_{s:(A')^\bullet \to A^\bullet \text{ qis and } (A')^\bullet \text{ bounded}} \mathrm{Hom}_{K^b(\mathcal{A})}((A')^\bullet, B^\bullet)$$

by Derived Categories, Lemma 05RW. Thus this is a countable set as a countable colimit of $\square$

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.